TS Physics 2015

Class 12 Physics,

Paper-II

Time allowed : 3hours **Maximum Marks : 60**

SECTION-A (10 × 2 = 20)

Note: i) Answer all the questions.

ii) Each question carries two marks.

iii) All are very short answer type questions.

1. Define power of a convex lens what is its unit.

Ans:- The power of a lens is defined as the reciprocal of the focal length. Lens power is measured in dioptres (D).

2. Distinguish between Ammeter and voltmeter.

Ans: **Ammeter:-**

1. It is a low resistance instrument.

2. Shunt resistance is very less.

3. It is always connected in series.

4. Resistance of an ideal ammeter is zero.

5. Its resistance is less than that of the galvanometer.

Voltmeter:

1. It is a high resistance instrument.

2. Series resistance is high.

3. It is always connected in parallel.

4. Resistance of an ideal voltmeter is infinity.

5. Its resistance is greater than that of voltmter.

3. Define magnetic inclination and the angle of dip.

Ans: Angle of dip is defined as Magnetic dip, dip angle, or magnetic inclination is the angle made with the horizontal by the Earth's magnetic field lines. This angle varies at different points on the Earth's surface

4. Classify the following material with regards to magnetism. (manganese, cobalt, nickel,bismuth, oxygen , copper)

Ans:

1. Diamagnetism

   manganese, bismuth, copper.

2. Paramagnetism

   manganese, oxygen.

3. Ferromagnetism

   cobalt, nickel.

5. What type of transformer is used in a 6v bed lamp ?

Ans:- In India, the household supply in is 220V−240VAC so to use a bed lamp having rating 6V a step downstep - downstep - down transformer will be required with a rectifierrectifier rectifier to convert 6V AC to 6V DC if the lamp is DC lamp.

6. What are the applications of microwaves ?

Ans: Microwaves are most commonly used in

- satellite communications
- radar signals, phones
- navigational applications
- medical treatments
- drying materials

7. What are the Cathode rays?

Ans: Cathode rays are streams of electrons observed in vacuum tubes. If an evacuated glass tube is equipped with two electrodes and a voltage is applied, glass behind the positive electrode is observed to glow, due to electrons emitted from the cathode

8. What is photoelectric effect?

Ans: The photoelectric effect is the emission of electrons or other free carriers when light hits a material. Electrons emitted in this manner can be called photoelectrons. This phenomenon is commonly studied in electronic physics and in fields of chemistry such as quantum chemistry and electrochemistry.

9. What is p-type semiconductor ? What are the majority

and minority charge carrier in it.

Ans : To make extrinsic semiconductors, when the impurity to be doped with intrinsic semiconductor is trivalent then the made semiconductor is called p-type semiconductor Boron and aluminum are some examples of trivalent impurity. Hole are majority charge carriers and electrons as minority charge carriers due to their deficiency.

10. Define modulation. Why is it necessary.

Ans: Modulation refers to the variation in strength, tone of individual's voice.

1. Modulation is necessary because due to modulation antenna size gets decrease.
2. Just because of modulation communication range increases.

SECTION- B (6 × 4 = 24)

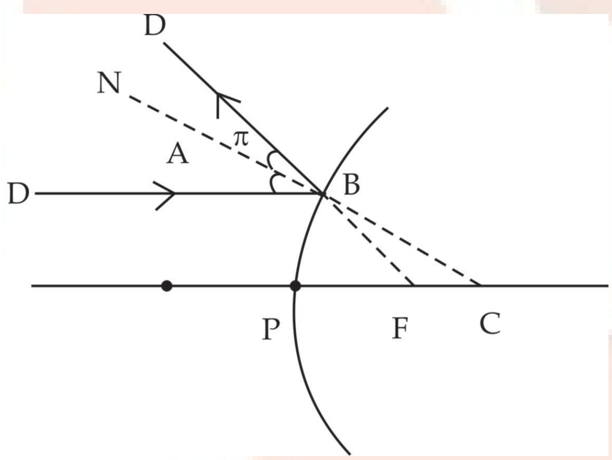
Note: i) Answer Any Six of the following questions.

ii) Each question carries four marks.

iii) All are short answer type questions

11. Define the focal length of a concave mirror . prove that the radius of curvature of a concave mirror is double its focal length.

Ans - Focal length (shown in red) is the distance between the center of a convex lens or a concave mirror and the focal point of the lens or mirror the point where parallel rays of light meet, or converge.



A ray of light AB traveling parallel to the principal axis PC is incident on a convex mirror at B. After reflection, it goes to Dand appear to be coming from the focus F.

The distance PF= focal length f.

The distance PC=radius of curvature R of the mirror.

Straight line NBCis the normal to the mirror at the point of incidence B.

$\angle ABN = \angle NBD$(Law of reflection, $\angle i = \angle r$)

$\angle CBF = \angle DBN$ (vertically opposite angles)

$\angle NBA = \angle BCF$(corresponding angles

$\Rightarrow \angle BCF = \angle CBF$

$\therefore$ ΔFBC is an isosceles triangle.

Hence, sides BF=FC

For a small aperture of the mirror, the point Bis very close to the point P,

$\Rightarrow BF = PF$ $\therefore PF = FC = 1/2PC$

$\Rightarrow f = 1/2R$

Thus, for a spherical mirror (both for a concave and for convex), the focal length is half of radius of curvature.

12. How do you determine the resolving power of your eye?

Ans: It is done by Retinoscopy. It is a instrument designed to see the reflexes of an intra molecular line. Light is flashed on dilated pupil and the lens reflex is observed.

13. Derive an expression for the intensity of the electric field at a point on the axial line of an electric dipole.

Ans : (a) For points on axial line

The axial line of a dipole is the line passing through the positive and negative charges of the electric dipole.

Consider a system of charges (-q and +q) separated by a distance 2a. Let 'P' be any point on an axis where the field intensity is to be determined.

Electric field at P (EB) due to +q

$$E_B = \frac{1}{4\pi\varepsilon_0}\frac{q}{(BP)^2} along\ BP$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2}$$

Electric field at P due to -q (EA)

$$E_A = \frac{1}{4\pi\varepsilon_0}\frac{1}{(AP)^2} along\ PA$$

$$\frac{1}{4\pi\varepsilon_0}\frac{1}{(r+a)^2}$$

Net field at P is given by

$$E_p = E_B - E_A$$

$$=\frac{1}{4\pi\varepsilon_0}\left[\frac{q}{(r-a)^2}-\frac{q}{(r+a)^2}\right]$$

Simplifying, we get

$$E_P=\frac{q}{4\pi\varepsilon_0}-\frac{4ar}{\left(r^2-a^2\right)^2}$$

$$E_P=\frac{2qr}{4\pi\varepsilon_0}-\frac{\partial}{\left(r^2-a^2\right)^2}$$

$$\left(2aq=p, K=\frac{1}{4\pi\varepsilon_0}\right)$$

$$or\ E_P=\frac{2kpr}{\left(r^2-a^2\right)^2} along\ BP$$

As a special case:

$$If\ 2a\,r,\ E_p=\frac{2kp}{r^3} along\ BP$$

14. Three capacitors of capicitances 2PF, 3PF, AND 4PF are connected in parallel.

a) what is the total capacitance of the combination ?

Ans - total capacitance = $C_1 + C_2 + C_3$

$$\rightarrow 2\text{pf} + 3\text{pf} + 4\text{pf}$$

$$\rightarrow 9\text{ pf}$$

b) Determine the total charge in the capacitor if the combination is connected tio a supply of 100V ?

Ans - charge q = cv

9×100

$\rightarrow$ 900 columb.

15. State and explain Biot-Savart-law .

Ans - biot Savart Law states that

The magnetic intensity dH at a point A due to current I flowing through a small element dl is

1. Directly proportional to current (I)
2. Directly proportional to the length of the element (dl)
3. Directly proportional to the sine of angle θ between the direction of current and the line joining the element dl from point A.
4. Inversely proportional to the square of the distance (x) of point A from the element dl.

$$dH = \frac{\mu_0 \mu_r}{4\pi} x\, Idl \sin\theta / x^2$$

$$dH = kx\, Idl \sin\theta / x^2$$

$$dH \propto Idl \sin\theta / x^2$$

where k is constant and depends on the magnetic properties of the medium.

$k = \mu_0 \mu_4 / 4\pi$

μ_0 = absolute permeability of air or vacuum and its value is 4 x 10–7 Wb/A–m

μr = relative permeability of the medium.

16. Describe the ways to which the eddy current are use to advantage.

Ans - eddy current brakes use the drag force created by eddy currents as a brake to slow or stop moving objects. Since there is no contact with a brake shoe or drum, there is no mechanical wear. However, an eddy current brake cannot provide a holding torque and so may be used in combination with mechanical brakes, for example, on overhead cranes.

Eddy current used as crack detection equipment, can be divided into high frequency instruments for finding surface breaking cracks in ferrous and non-ferrous materials and low frequency instruments for finding cracks in non-ferrous materials.

17. What are the limitations of Bohr's theory of hydrogen atom?

Ans: Limitations of bohr's theory are:

1. Bohr's theory of hydrogen atom does not explain the Zeeman effect.
2. Bohr's model of hydrogen atom is very restricted in terms of size.

3. It does not explain the relative intensities of spectral lines.
4. It couldn't able to explain the reason, When speed up electrons don't emit the electromagnetic radiations.

18. Distinguish between half wave and full wave rectfiers.

Ans: - Half wave rectifier conducts current only during positive half cycle of the input whereas full wave rectifier conducts both positive as well as negative half cycle of the input.

1. Output frequency of half wave rectifier is equal to the frequency of input whereas in full wave rectifier output frequency is twice of the input.
2. Only one diode is required for half wave rectifier whereas two diode are required for full wave rectifier.
3. For half wave rectifier, there is no need for center tapped transformer whereas center tapped transformer is required for full wave rectifier.
4. Efficiency of half wave rectifier is 40.6% whereas efficiency of full wave rectifier is 81.2%

SECTION –C (2 × 18 =16)

Note: i) Answer ANY TWO of the following questions.

ii) Each question carries Eight marks.

iii) All are long answer type questions

19. Explain the formation of stationary waves in air column enclosed in open pipe.

Derive the equation for the frequencies of the harmonics produced a closed organ pipe 70 cm long is sounded. if the velocity of sound is 311m/s, what is the fundamental frequency of vibration of the air columb ?

Ans: The air at the closed end is not free to vibrate. Hence a node is formed at the closed end. Air at the open end is free to vibrate with maximum amplitude and antinode is formed at the open end. This is the simplest mode of vibration of an air column closed at one end and is called the fundamental mode.

Each harmonic frequency (fn) is given by the equation fn = n. ... Rearranging this equation leads to f1 = fn / n. So using the fifth harmonic frequency (f5) or the third harmonic frequency (f3), the first harmonic frequency can be calculated.

The fundamental frequency of vibration of the air column is 118.2 Hz.

Explanation:

Given that,

length l = 70 cm

Velocity of sound = 331 m/s

We know that,

The fundamental frequency of closed organ pipe is

f=v/4l .....(I)

Where, v = velocity of sound

l = length

Put the value of equation (I)

f=331/4×70×10^-2

f=118.2 Hz

Hence, The fundamental frequency of vibration of the air column is 118.2 Hz.

20. State the working principle of potentiometer. explain with the help of circuit diagram how the emf of two primary cells which balances against a length of 180 cm of the potentiometer wire.

Ans: potentiometer works on the principle of galvanometer.

when we keep key (k1) closed and (k2) open ,let the null point found be Iaj1

E1= KIaj1....... (1)

when we keeep k1 open and k2 closed , let null point obtained by IaJ2.

E2=KIaj2............(2)

(1)/(2)

therefore E1/E2= I1/I2

where E1 and E2 are the two EMF of the given two cell.

21. Example the principle and working of nuclear reactor with the help of labelled diagram .

Ans: Nuclear reactors operate on the principle of nuclear fission, the process in which a heavy atomic nucleus splits into two smaller fragments. The nuclear fragments are in very excited states and emit neutrons, other subatomic particles, and photons.

Nuclear reactors are the heart of a nuclear power plant. They contain and control nuclear chain reactions that produce heat through a physical process called fission. That heat is used to make steam that spins a turbine to create electricity.